

Loan Portfolio & Credit Risk Analytics

Internship Project Report
Python • SQL • Power BI

Dataset	10,000 public Lending Club-style loan records
Objective	Understand portfolio composition, borrower characteristics, loan performance, and credit-risk patterns.
Approach	Data cleaning, feature engineering, exploratory analysis, SQL analysis, and dashboard development.
Currency	United States dollars (USD)

Executive Summary

This project presents an end-to-end analytical workflow for a loan portfolio. The dataset was prepared in Python, analyzed through descriptive statistics and visualizations, queried with SQL, and presented through a Power BI dashboard.

The analysis covered 10,000 loans with a total loan amount of approximately **\$163.62 million** and an outstanding balance of approximately **\$144.59 million**. The average loan amount was approximately **\$16.36 thousand**, the average interest rate was **12.43%**, and total interest paid was approximately **\$6.00 million**.

Debt consolidation was the largest loan-purpose segment by total loan amount, while credit grades A, B, and C represented most of the records. Most loans were classified as Current in the supplied sample.

Risk categories used in this report were created specifically for this project. They are analytical groupings and should not be interpreted as official lender classifications.

1. Project Objectives

- Measure the overall size and composition of the loan portfolio.
- Analyze loans by purpose, credit grade, term, and loan status.
- Study annual income, debt-to-income ratio, credit utilization, and credit history variables.
- Create project-defined risk categories from loan-status information.
- Compare risk patterns across credit grades, DTI groups, and loan purposes.
- Use SQL to reproduce portfolio summaries and grouped analyses.
- Develop an interactive Power BI dashboard.

2. Dataset and Tools

The project uses a public Lending Club-style sample containing 10,000 loan records and 56 original columns. Fields include loan amount, interest rate, installment, credit grade, loan status, loan purpose, income, debt-to-income ratio, employment length, homeownership, and credit-history measures.

Tool	Purpose
Python / Google Colab	Data loading, cleaning, feature engineering, and analysis
Pandas and NumPy	Data manipulation and numerical processing
Matplotlib and Seaborn	Exploratory visualizations
DuckDB / SQL	Grouped portfolio queries
Power BI	Interactive dashboard and filtering

3. Data Preparation

- Selected 24 relevant original columns for the main analysis.
- Excluded highly incomplete joint-income variables from the main analysis.
- Filled missing numerical values using median imputation.
- Filled missing categorical values using "Unknown".
- Created credit utilization, DTI groups, and income groups.
- Created a project-defined risk category from loan status.
- Converted issue month into a date field.
- Validated the final dataset for missing values and duplicates.

The final analytical dataset contained **10,000 rows and 30 columns**, with no remaining missing values and no duplicate rows.

4. Portfolio Analysis

Metric	Result
Total loans	10,000
Total loan amount	\$163.62 million
Outstanding balance	\$144.59 million
Average loan amount	\$16.36 thousand
Average interest rate	12.43%
Total interest paid	\$6.00 million

5. Main Analytical Observations

- Debt consolidation was the largest segment by total loan amount.
- Grades A, B, and C accounted for most portfolio records.
- 36-month loans were more common than 60-month loans.
- Mortgage and rent were the most common homeownership categories.
- Most records were classified as Current, with smaller groups in fully paid, grace-period, late, and charged-off statuses.
- The 10–20 DTI group was the largest DTI segment.
- Issue-month data covered January, February, and March 2018 only.

6. Credit Risk Analysis

For grouped analysis, loan statuses were mapped into the following project-defined categories:

Category	Included statuses
High Risk	Charged Off and Default
Medium Risk	Late (31–120 days) and Late (16–30 days)
No Current Problem	All remaining statuses

Under this definition, the dataset contained **7 High Risk**, **104 Medium Risk**, and **9,889 No Current Problem** records. These figures describe the selected sample and classification rule; they are not a formal default-rate estimate for the wider lending market.

7. Power BI Dashboard

The Power BI dashboard combines portfolio-level KPIs with charts for loan purpose, loan status, credit grade, risk category, and monthly loan issuance. Slicers allow exploration by loan purpose and credit grade.

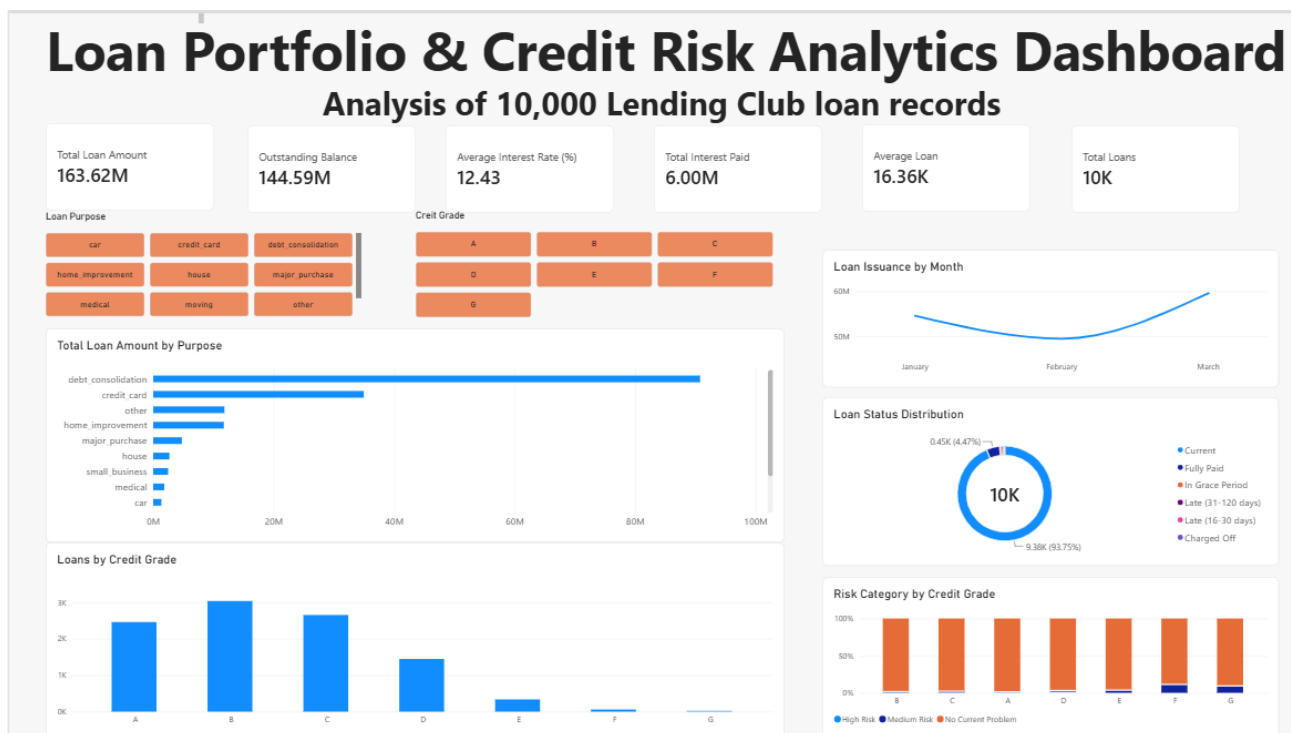


Figure 1. Power BI dashboard showing portfolio KPIs, loan-purpose distribution, loan status, credit grades, risk by grade, and monthly issuance.

Dashboard Components

- Six KPI cards summarize total loan amount, outstanding balance, average interest rate, total interest paid, average loan amount, and total loans.
- A bar chart displays total loan amount by purpose.
- A donut chart presents loan status distribution.
- A column chart shows the number of loans by credit grade.
- A stacked chart compares project-defined risk categories within each credit grade.
- A line chart displays loan issuance for the three available issue months.

8. SQL Analysis

DuckDB was used in Google Colab to load the cleaned CSV file into a table named **loans**. SQL queries reproduced important portfolio summaries and grouped analyses.

- Overall portfolio summary
- Portfolio totals by loan purpose
- Portfolio totals by credit grade
- Loan status distribution
- Risk category distribution
- Risk percentages within each credit grade using a window function
- Risk percentages within each DTI group using a window function
- Monthly portfolio summary

9. Business Insights

- Portfolio monitoring should track loan-purpose concentration because a large share of exposure is associated with debt consolidation.
- Credit-grade and DTI comparisons can be used as monitoring dimensions for portfolio segmentation.
- Loan status distribution should be reviewed regularly to identify changes in late or charged-off records.
- The dashboard supports quick exploration through loan-purpose and credit-grade filters.
- Monthly issuance should be interpreted as a short snapshot because only three issue months are available.

10. Limitations

- The dataset is a 10,000-record sample and may not represent a complete lending portfolio.
- Monthly analysis covers only January–March 2018.
- Risk categories are project-defined analytical groupings.
- The analysis identifies associations and patterns rather than causation.
- Some credit-grade groups contain relatively few records.
- No machine-learning prediction model was included because the project focuses on descriptive analytics.

11. Conclusion

This project demonstrates a complete loan portfolio analytics workflow, from raw-data inspection and preparation through feature engineering, exploratory analysis, SQL queries, and Power BI visualization. The notebook, cleaned dataset, SQL file, dashboard, and report provide a reproducible foundation for understanding portfolio composition and credit-risk patterns.